

Zebra Mussel Invasion Across the Great Lake Basin

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Introduction

Zebra Mussels originate from the Caspian and Black Seas. Through shipping they were introduced into the Great Lakes region in the late 1980s.

Due to their invasive nature, they have overrun the native mussel population and have other effects on algae and pipes.

In recent years, they have started to compete with themselves, but have also spread outside of the Great Lakes through the Mississippi and Ohio rivers up to out to the Minnesota/North Dakota border.

Hotspots

- Due to the reporting nature, the hotspots are likely skewed at least to an extent to areas with more swimming (seen with various inland lakes).
- Unsurprisingly, the reports trace the major waterways of the Mississippi and Ohio rivers.
- The early hotspots from 1988 to 1990 are focused on major port cities such as Duluth, Cleveland, Toledo and also the Niagara River.
- Throughout the whole analysis period, almost 25% of reports were within 25 miles of a major port.
 - In the first five years of the period (1988-1993), it's about 75% of reports near a major port.
 - Of those, 37% are near the Chicago/Gary ports.

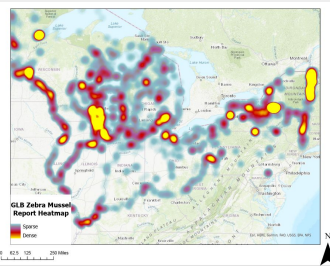


Figure 1: GLB Zebra Mussels Heatmap

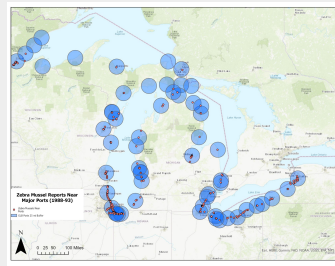


Figure 2: GLB Zebra Mussels Near Ports 1988-1993

Data and Methods

- Zebra mussel reports from GLANSIS (1988-2025).
- Great Lakes ports dataset from the Great Lakes Commission.
- Heatmap symbology used to show density of reports.
- Pairwise buffer ArcGIS tool to show reports near ports.
- Point to raster ArcGIS tool to show spread by year.

Top 5 Ports with Most Reports

1. Chicago-Calumet - 73
2. Milwaukee - 59
3. Indiana Harbor (Gary, IN) - 45
4. Waukegan - 45
5. Buffington Harbor (Gary, IN) - 43

Top 5 HUC8 with Most Reports

1. Ondeddia (NY) - 725
2. Lake Champlain (NY) - 665
3. Lake Michigan - 440
4. Upper Fox (IL) - 198
5. Lake Erie - 183

Spread

- The spread early on was throughout the Great Lakes, Mississippi River, and East NY State.
- More recently they have spread further into Minnesota and N. Michigan.

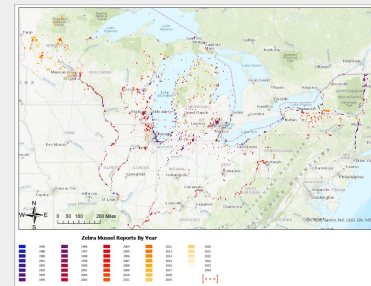


Figure 3: GLB Zebra Mussels Spread by Year

Conclusion

- Within areas already invaded, they will likely have to starve themselves out.
- The extent of shipping throughout the Great Lake Basin has made the region very vulnerable to aquatic invasive species.
 - With this, connected major waterways will allow zebra mussels to spread much further.

Sources

- U.S Department of the Interior. (n.d.). *Invasive zebra mussels* (U.S National Park Service). National Parks Service. <https://www.nps.gov/articles/zebra-mussels.htm>
- Glansis Map Explorer: NOAA Great Lakes Environmental Research Laboratory - Ann Arbor, MI, USA. (n.d.-a). <https://www.glerl.noaa.gov/glansis/mapExplorer.php>
- GLC Admin. (2023, October 18). *Great Lakes Ports*. ArcGIS. <https://arcgis.com/home/item.html?id=61a19b7f0bca48659d5ed710eeaba339>